

ST MARY RESERVOIR, AB, Canada

WMO: 718370

Lat: 49.3614N

Lon: 113.1100W

Elev: 1128

StdP: 88.49

Time Zone: -7.00 (NAM)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.2	-24.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	30.1	N/A	28.1	N/A	26.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	-32.2	32.7	2.5	2.4	-34.0	34.5	-35.5	35.8	-36.9	37.2	-38.7	38.9
			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-4.9	-6.7	-1.1	4.5	10.1	14.4	18.1	17.6	13.0	6.2	-0.2	-6.4		
	DBStd	10.67	7.83	8.53	7.18	5.10	4.37	3.36	3.15	3.33	4.96	5.50	7.14	8.05		
	HDD10.0	2489	463	467	345	173	54	4	0	1	27	140	308	507		
	HDD18.3	4783	722	701	603	414	256	123	43	54	168	378	556	766		
	CDD10.0	830	0	0	0	9	57	137	251	237	116	21	3	0		
	CDD18.3	80	0	0	0	0	1	6	35	31	7	0	0	0		
	CDH23.3	1256	0	0	0	4	29	93	496	475	156	3	0	0		
	CDH26.7	344	0	0	0	0	1	14	151	145	33	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	548	27	24	34	49	76	107	47	46	47	33	31	28		
	PrecMax	844	43	49	64	94	166	323	173	118	135	79	61	59		
	PrecMin	343	8	6	12	17	9	24	3	3	5	6	10	9		
	PrecStd	120	9	11	13	17	38	67	38	28	29	19	13	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.9	10.9	13.6	23.4	26.1	29.0	33.0	32.6	29.8	22.7	17.4	10.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	7.6	7.4	11.6	18.4	23.9	25.8	30.5	30.1	27.4	19.2	13.2	7.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	5.9	5.5	9.6	15.3	21.5	23.9	28.1	28.1	24.9	16.4	10.6	5.2		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	4.5	3.8	7.4	12.6	18.9	22.0	26.1	26.0	22.0	13.8	8.4	3.5		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	MDBR		9.0	9.5	9.8	10.7	11.4	11.3	13.4	13.9	12.5	10.4	8.8	8.6		
	5% DB	MCDBR	9.1	9.9	12.0	15.2	15.9	14.7	16.7	17.7	17.4	13.6	11.2	9.1		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.244	0.256	0.277	0.318	0.328	0.324	0.329	0.345	0.309	0.277	0.253	0.229		
	taud		2.429	2.446	2.480	2.383	2.415	2.470	2.461	2.395	2.473	2.542	2.501	2.437		
	Ebn at Noon		842	915	948	931	931	934	924	891	897	879	826	817		
	Edh at Noon		60	76	88	109	111	105	105	107	89	69	56	52		
(44) All-Sky Solar Radiation	RadAvg		1.24	2.25	3.48	4.82	5.74	6.41	7.10	5.66	4.02	2.50	1.39	0.99		
	RadStd		0.10	0.17	0.30	0.32	0.50	0.48	0.38	0.35	0.33	0.25	0.11	0.10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page